



**NORTH
SYDNEY
GIRLS HIGH
SCHOOL**

2020

**Preliminary
Mathematics
Assessment
Task 2**

Preliminary Mathematics Advanced

General Instructions

- Reading Time – 2 minutes
- Working Time – 55 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 6 – 9, show relevant mathematical reasoning and/or calculations

Total marks – 50

Section I – 5 marks (pages 2 – 3)

- Attempt Questions 1 – 5
- Allow about 6 minutes for this section

Section II – 45 marks (pages 5 – 20)

- Attempt Questions 6 – 9
- Allow about 49 minutes for this section

NAME: _____

TEACHER: _____

Question	1 – 5	6	7	8	9	Total
Algebra and Arithmetic	/2	/11	/2			/15
Functions	/3	/2	/9		/2	/16
Trigonometry				/11	/8	/19
	/5	/13	/11	/11	/10	/50

Section I

5 marks

Attempt Questions 1-5

Allow about 6 minutes for this section

Use the multiple choice answer sheet for Questions 1-5.

1 What is the value of $\frac{5\sqrt{\pi}}{\cos 28^\circ}$, correct to 2 significant figures?

- A. 10.04
- B. 10.03
- C. 1.04×10^1
- D. 1.0×10^1

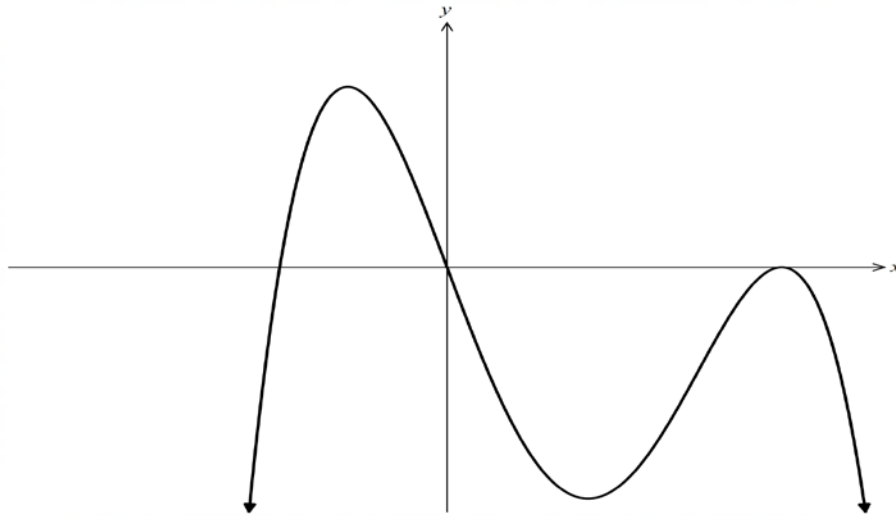
2 What is the simplest expression for $5^x \times 5^{2x}$?

- A. 5^{2x^2}
- B. 5^{3x}
- C. 25^{2x^2}
- D. 25^{3x}

3 What is the domain of $y = \sqrt{x-3} - \sqrt{7-x}$?

- A. $(-\infty, 3) \cup (7, \infty)$
- B. $(-\infty, 3] \cup [7, \infty)$
- C. $[3, 7]$
- D. $(3, 7)$

- 4 Consider the graph below.



Which of the following equations could represent the graph?

- A. $y = -3x(x-4)^2(x+2)$
- B. $y = 3x(x-4)^2(x+2)$
- C. $y = -3x(x+4)^2(x-2)$
- D. $y = 3x(x+4)^2(x-2)$
- 5 If $f(x-1) = x^2 - 2x + 3$, then which of the following is an expression for $f(x)$?

- A. $x^2 - 2$
- B. $x^2 + 2$
- C. $x^2 - 2x + 2$
- D. $x^2 - 2x + 4$

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Section II

50 marks

Attempt Questions 6 to 9.

Allow about 49 minutes for this section.

In questions 6 to 9, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (13 marks)

(a) Given that $\log_a b = 4.25$ and $\log_a c = 0.75$, find the value of:

(i) $\log_a \left(\frac{c}{b} \right)$ 1

(ii) $\log_a (bc)^3$. 1

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(b) Solve $|4 - 3x| = 7$ 2

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Question 6 continues on Page 6

Question 6 (continued)

- (c) Write $\frac{1+\sqrt{xy}}{x\sqrt{x}}$ in simplest form with a rational denominator, given $x, y > 0$ 2

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- (d) Solve $2x^2 - 2x - 1 = 0$. 2

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Question 6 continues on Page 7

Question 6 (continued)

(e) Solve the following pair of simultaneous equations:

3

$$7^{y-2x} = \frac{1}{7}$$

$$x^2 - 2x + y = 0$$

[illegible]

Question 6 continues on Page 8

(f) The resistance to the motion of a train R is the sum of two parts, one is a constant of 300 Newtons and the other is directly proportional to square of its speed, v m/s. If the total resistance is 350 Newtons when the speed of the train is 30 m/s, calculate the total resistance when the speed is 36 m/s.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Question 7 (11 marks)

(a) Solve $12 + 5x - 2x^2 \leq 0$

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(b) The equation of a circle is $x^2 + y^2 - 2x + 4y - 20 = 0$. Find the circle's centre and radius.

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Question 7 continues on Page 10

(c) A function is defined as:

(i) Evaluate $f(-3) + f(2) - f(0)$. **2**

[illegible]

- 10 -

Question 7 (continued)

(d) Let $f(x) = 3 + \sqrt{4 - x}$ and $g(x) = x^2$

- (i) find an expression for $f(g(x))$ **1**
- (ii) find the domain of $f(g(x))$ **1**
- (iii) sketch $y = f(g(x))$. **2**

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

This image shows a full page of a handwriting practice sheet. It consists of multiple horizontal rows, each defined by two parallel dotted lines. The rows are evenly spaced and extend across the entire width of the page, providing a guide for letter height and placement. There is no text or other markings on the page.

Question 8 (11 marks)

- (a) Express 495° in radians as an exact value. **1**

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- (b) Find the exact value of $\csc\left(-\frac{4\pi}{3}\right)$. **1**

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- (c) If $\cos\alpha = \frac{2}{5}$ and α is a reflex angle, find the exact value of $\tan\alpha$. **2**

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Question 8 continues on Page 14

Question 8 (continued)

(d) Show that $\frac{1}{\sin \alpha} - \frac{1}{\sin^2 \alpha} = \frac{\sin \alpha - 1}{(1 + \cos \alpha)(1 - \cos \alpha)}$. 2

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(e) Sketch $y = 2 + \sin\left(\frac{\pi x}{3}\right)$ for $0 \leq x \leq 12$, showing all important features. 2



Question 8 continues on Page 15

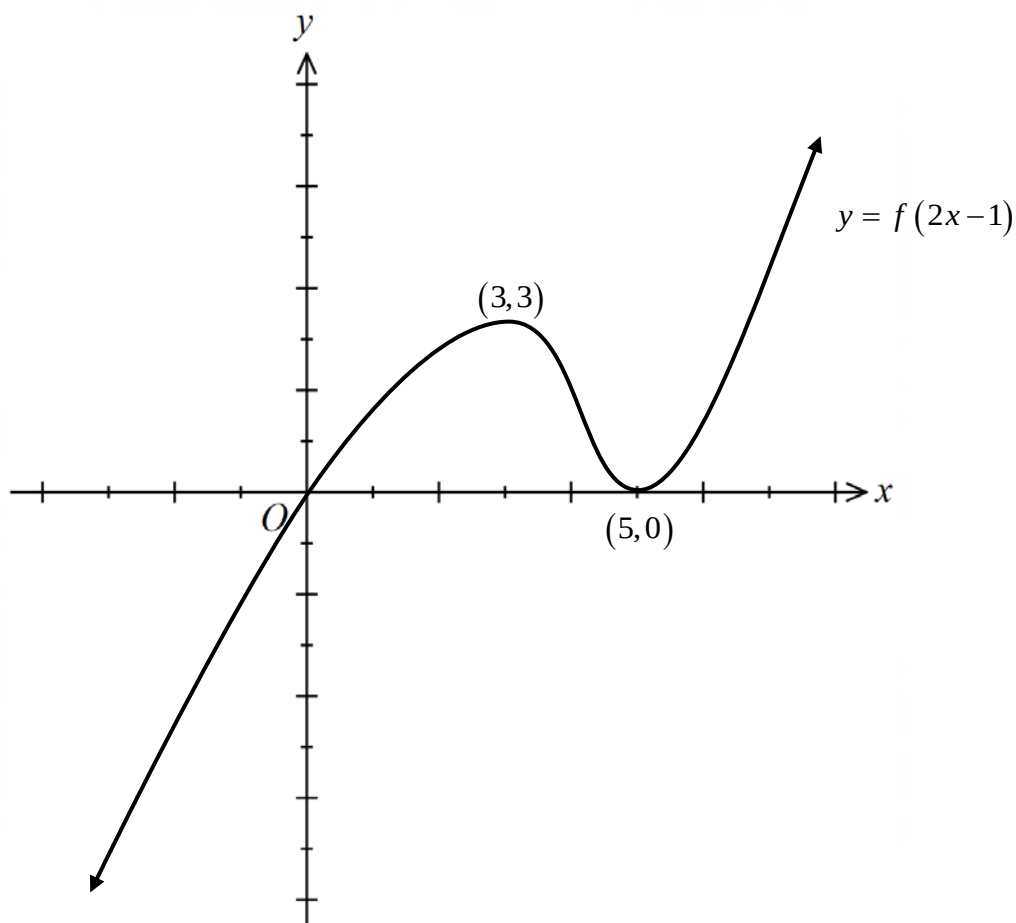
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Question 9 (10 marks)

- (a) The graph of the curve $y = f(2x - 1)$ is shown below.

2

The graph passes through the origin and points $(3, 3)$ and $(5, 0)$ respectively as shown.



By using the graph, or otherwise:

- (a) Find the number of solutions to the equation $f(2x - 1) - |x| = 0$.

1

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- (b) Solve $f(x) = 0$

1

.....

Question 9 continues on Page 18

(b) $\triangle ABC$ is an equilateral triangle with sides of length 6 m.

A diagram of a triangle ABC with vertex A at the top and base BC at the bottom. A dashed line segment connects vertex A to the base BC . The base BC is divided into two segments, BX and CY , which are shaded blue. A right-angle symbol is shown at the intersection of the dashed line and the base BC .

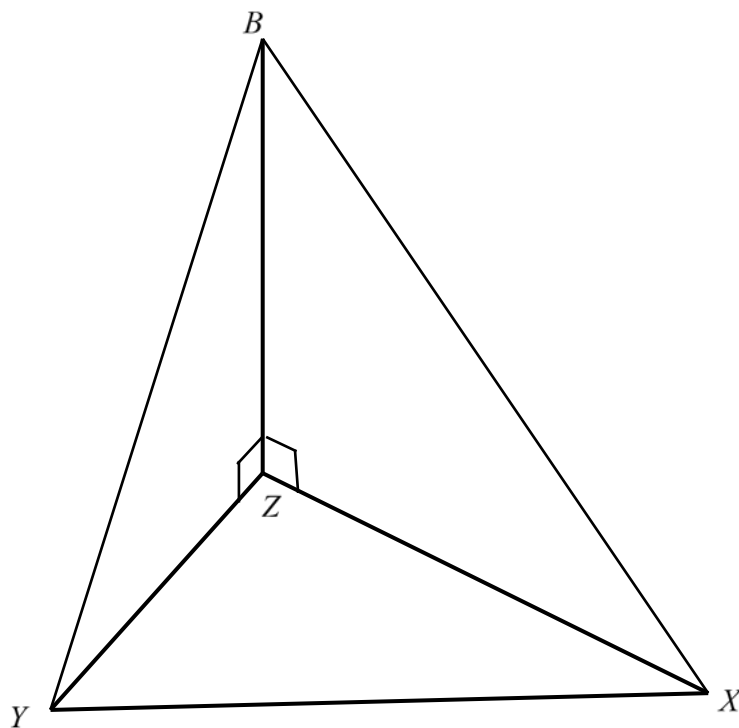
- (i) Show that the radius of the arc is $3\sqrt{3}$ m. **1**
- (ii) Find the exact area of the shaded region. **3**

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Question 9 continues on Page 19

Question 9 (continued)

- (c) The angle of elevation of a hot air balloon B from a point on the ground X is 29° and the bearing is 323° .
From point Y 4000 m due west of X , the bearing of the balloon is 049° .



- | | |
|--|---|
| (i) Draw $\triangle XYZ$ showing the bearings. | 1 |
| (ii) Explain why $\angle YZX = 86^\circ$. | 1 |
| (iii) Find the altitude of the balloon. | 2 |

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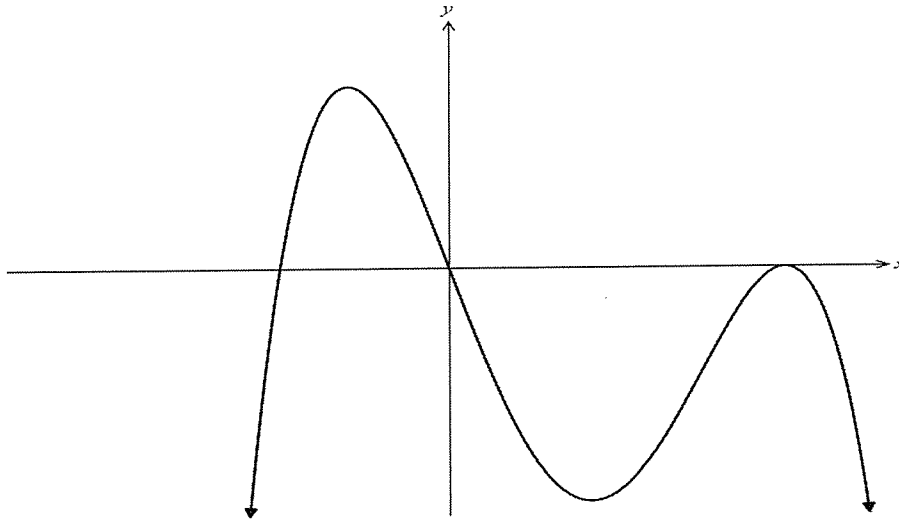
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Which of the following equations could represent the graph?

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- B. $y = 3x(x-4)^2(x+2)$
- C. $y = -3x(x+4)^2(x-2)$
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- D. $x^2 - 2x + 4$

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Section II

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Question 6 (13 marks)

(a) Given that $\log_a b = 4.25$ and $\log_a c = 0.75$, find the value of:

(i) $\log_a \left(\frac{c}{b} \right)$

1

(ii) $\log_a (bc)^3$

1

(i) $\log_a c - \log_a b = 0.75 - 4.25$
 $= -3.5$

(ii) $3 \log_a bc = 3(\log_a b + \log_a c)$
 $= 3(4.25 + 0.75)$
 $= 15$

(b) Solve $|4 - 3x| = 7$

2

$4 - 3x = 7$ or $-4 + 3x = 7$
 $3x = -3$ $3x = 11$
 $x = -1$ $x = \frac{11}{3}$

Question 6 continues on Page 6

Question 6 (continued)

- (c) Write $\frac{1+\sqrt{xy}}{x\sqrt{x}}$ in simplest form with a rational denominator, given $x, y > 0$

2

$$\begin{aligned} & \frac{(1 + \sqrt{xy})}{x\sqrt{x}} \times \frac{\sqrt{x}}{\sqrt{x}} \\ &= \frac{\sqrt{x} + \sqrt{x} \cdot \sqrt{xy}}{x \cdot x} \\ &= \frac{\sqrt{x} + x\sqrt{y}}{x^2} \end{aligned}$$

- (d) Solve $2x^2 - 2x - 1 = 0$.

2

$$\begin{aligned} x &= \frac{2 \pm \sqrt{(-2)^2 - 4(2)(-1)}}{4} \\ &= \frac{2 \pm \sqrt{12}}{4} \\ &= \frac{2 \pm 2\sqrt{3}}{4} \\ &= \frac{1 \pm \sqrt{3}}{2} \end{aligned}$$

Question 6 continues on Page 7

Question 6 (continued)

(e) Solve the following pair of simultaneous equations:

3

$$7^{y-2x} = \frac{1}{7} = 7^{-1} \quad \text{--- ①}$$

$$x^2 - 2x + y = 0$$

From ① $\Rightarrow y - 2x = -1$

$$y = 2x - 1$$

From ② $\Rightarrow$

$$y = -x^2 + 2x$$

$$-x^2 + 2x = 2x - 1$$

$$x^2 = 1$$

$$x = 1 \text{ or } x = -1$$

$$x = -1$$

$$y = 2(1) - 1$$

$$y = 2(-1) - 1$$

$$y = 1$$

$$y = -3$$

$\therefore$ solutions are $x=1, y=1$ or $x=-1, y=-3$

Question 6 continues on Page 8

Question 6 (continued)

- (f) The resistance to the motion of a train R is the sum of two parts, one is a constant of 300 Newtons and the other is directly proportional to square of its speed, v m/s.

2

If the total resistance is 350 Newtons when the speed of the train is 30 m/s, calculate the total resistance when the speed is 36 m/s.

$$R = 300 + kV^2$$

find k : when $R = 350$, $V = 30$

$$350 = 300 + k \times 30^2$$

$$50 = 900k$$

$$k = \frac{50}{900} = \frac{1}{18}$$

$$\therefore R = 300 + \frac{V^2}{18}$$

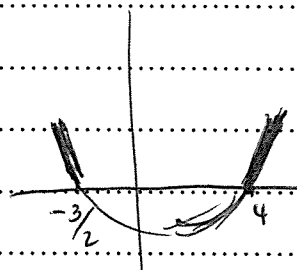
when $V = 36$, $R = 300 + \frac{36^2}{18}$

$$= 372 \text{ newtons}$$

Question 7 (11 marks)

(a) Solve $12 + 5x - 2x^2 \leq 0$

$$\begin{aligned} & \dots\dots\dots 2x^2 - 5x - 12 \geq 0 \dots\dots\dots \text{or} \dots\dots\dots (3 + 2x)(4 - x) \leq 0 \\ & \dots\dots\dots \begin{matrix} 2x^2 & & -5x & & -12 \\ 2x & & -4 & & 3 \end{matrix} \\ & \dots\dots\dots (x - 4)(2x + 3) \geq 0 \\ & \dots\dots\dots x \geq 4 \text{ or } x \leq -\frac{3}{2} \end{aligned}$$



(b) The equation of a circle is $x^2 + y^2 - 2x + 4y - 20 = 0$. Find the circle's centre and radius.

$$\begin{aligned} & \dots\dots\dots x^2 + 2x + 1 + y^2 + 4y + 4 = 20 + 1 + 4 \\ & \dots\dots\dots (x - 1)^2 + (y + 2)^2 = 25 \\ & \dots\dots\dots \text{radius} = 5 \\ & \dots\dots\dots \text{centre} = (1, -2) \end{aligned}$$

Question 7 continues on Page 10

Question 7 (continued)

(c) A function is defined as:

$$f(x) = \begin{cases} 5x - 4 & \text{when } x \leq 0 \\ -x^2 & \text{when } x > 0 \end{cases}$$

(i) Evaluate $f(-3) + f(2) - f(0)$.

2

(ii) Write an expression for $f(a^2 + 2)$.

1

$$\begin{aligned} \text{(i)} \quad f(-3) &= 5(-3) - 4 = -19 \\ f(2) &= -(2)^2 = -4 \\ f(0) &= -4 \end{aligned}$$

$$\begin{aligned} f(-3) + f(2) - f(0) &= -19 - 4 + 4 \\ &= -19 \end{aligned}$$

$$\text{(ii)} \quad f(a^2 + 2) = -(a^2 + 2)^2 \text{ or } -a^4 - 4a^2 - 4$$

$$\text{as } a^2 + 2 \geq 0$$

Question 7 continues on Page 11

Question 7 (continued)

(d) Let $f(x) = 3 + \sqrt{4-x}$ and $g(x) = x^2$

(i) find an expression for $f(g(x))$

1

(ii) find the domain of $f(g(x))$

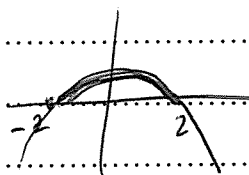
1

(iii) sketch $y = f(g(x))$.

2

(i) $f(g(x)) = f(x^2)$
 $= 3 + \sqrt{4 - x^2}$

(ii) $4 - x^2 \geq 0$
 $(2-x)(2+x) \geq 0$

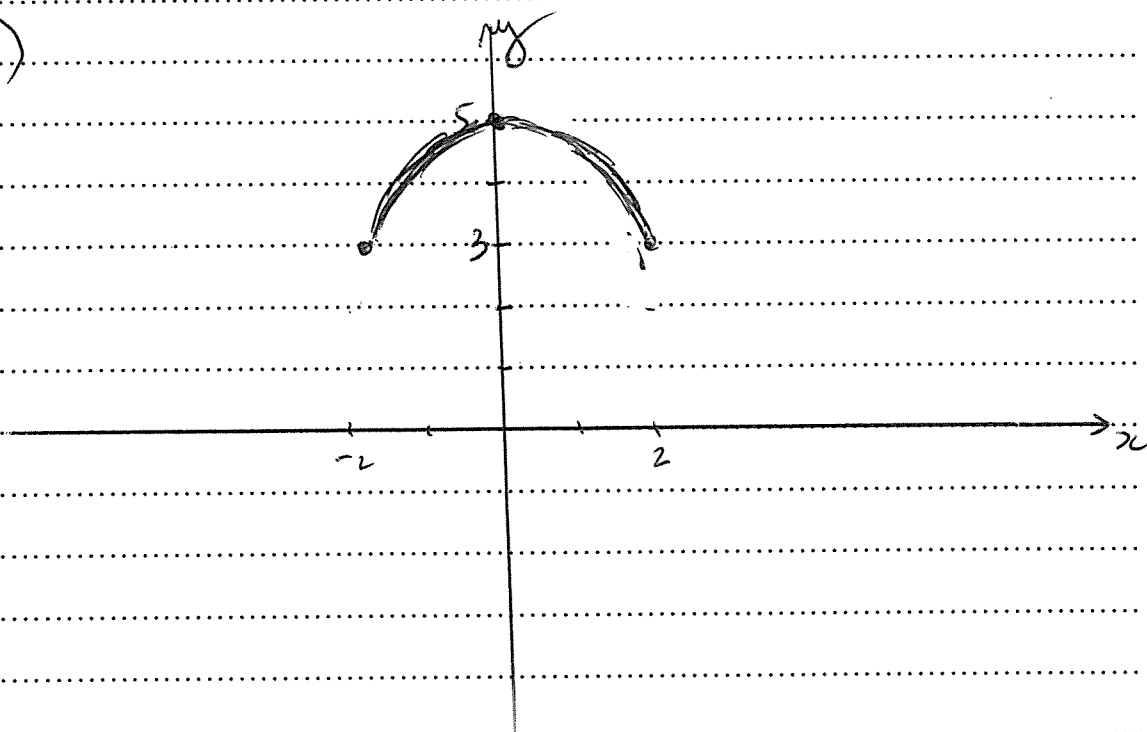


$-2 \leq x \leq 2$

and Domain of $g(x) = x^2 : -\infty < x < \infty$

$\therefore$ Domain of $f(g(x)) : -2 \leq x \leq 2$

(iii)



This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

Question 8 (11 marks)

- (a) Express 495° in radians as an exact value.

1

$$495^\circ \times \frac{\pi}{180} = \frac{11\pi}{4}$$

- (b) Find the exact value of $\csc\left(-\frac{4\pi}{3}\right)$.

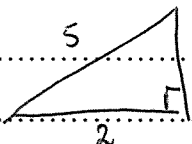
1

$$= \csc\left(\pi - \frac{\pi}{3}\right)$$

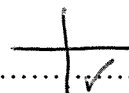
$$= \csc \frac{\pi}{3} = \frac{2}{\sqrt{3}}$$

- (c) If $\cos \alpha = \frac{2}{5}$ and α is a reflex angle, find the exact value of $\tan \alpha$.

2



$$\sqrt{25 - 4} = \sqrt{21}$$



$$\tan \alpha = -\frac{\sqrt{21}}{2}$$

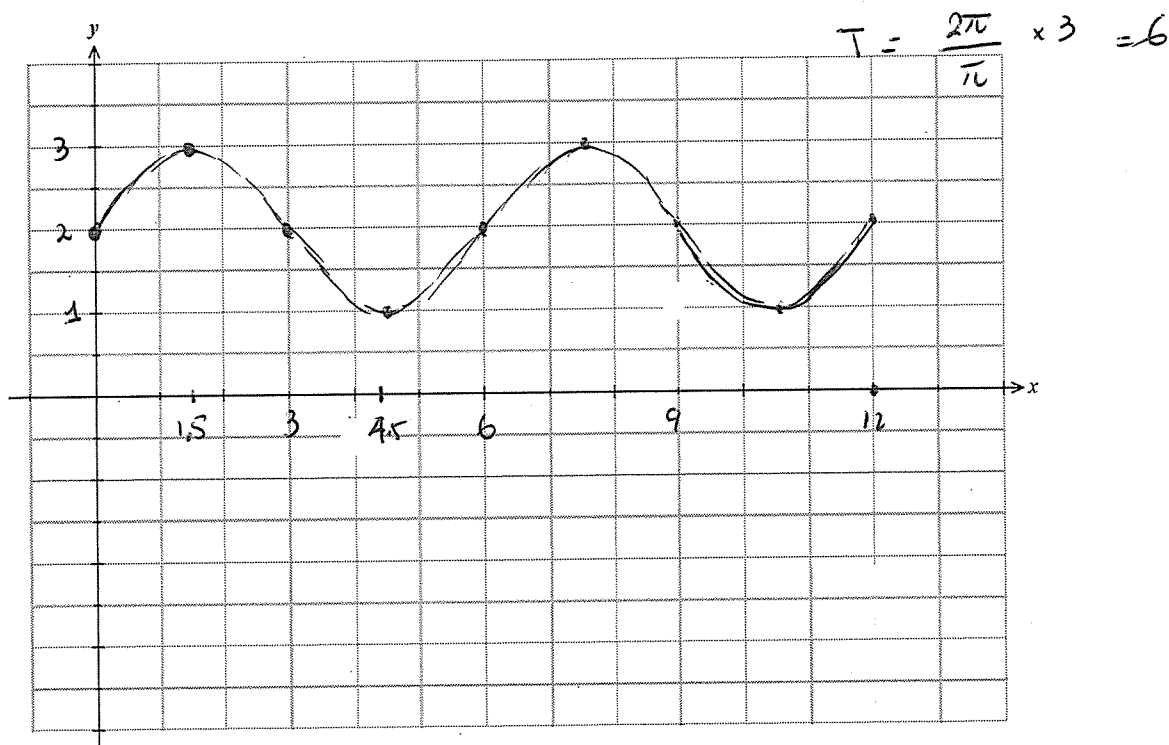
Question 8 continues on Page 14

Question 8 (continued)

- (d) Show that $\frac{1}{\sin \alpha} - \frac{1}{\sin^2 \alpha} = \frac{\sin \alpha - 1}{(1 + \cos \alpha)(1 - \cos \alpha)}$. 2

$$\begin{aligned}
 \text{LHS} &= \frac{1}{\sin \alpha} \times \frac{\sin \alpha}{\sin \alpha} = \frac{1}{\sin^2 \alpha} \\
 &= \frac{\sin \alpha - 1}{\sin^2 \alpha} \\
 &= \frac{\sin \alpha - 1}{1 - \cos^2 \alpha} \\
 &= \frac{\sin \alpha - 1}{(1 - \cos \alpha)(1 + \cos \alpha)} \\
 &= \text{RHS}
 \end{aligned}$$

- (e) Sketch $y = 2 + \sin\left(\frac{\pi x}{3}\right)$ for $0 \leq x \leq 12$, showing all important features. 2



Question 8 continues on Page 15

Question 8 (continued)

- (f) Solve $4\sin^2 x + 7\cos x = 4$, for $0^\circ \leq x \leq 360^\circ$.
Give your answers correct to the nearest minute.

3

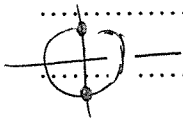
$$4\sin^2 x + 7\cos x - 4 = 0$$

$$4(1 - \cos^2 x) + 7\cos x - 4 = 0$$

$$4 - 4\cos^2 x + 7\cos x - 4 = 0$$

$$-4\cos^2 x + 7\cos x = 0$$

$$\cos x [7 - 4\cos x] = 0$$



$$\cos x = 0$$

$$x = 90^\circ, 270^\circ$$

$$4\cos x = 7$$

$$\cos x = \frac{7}{4}$$

↑

No solution

$$\text{as } -1 \leq \cos x \leq 1$$

[illegible]

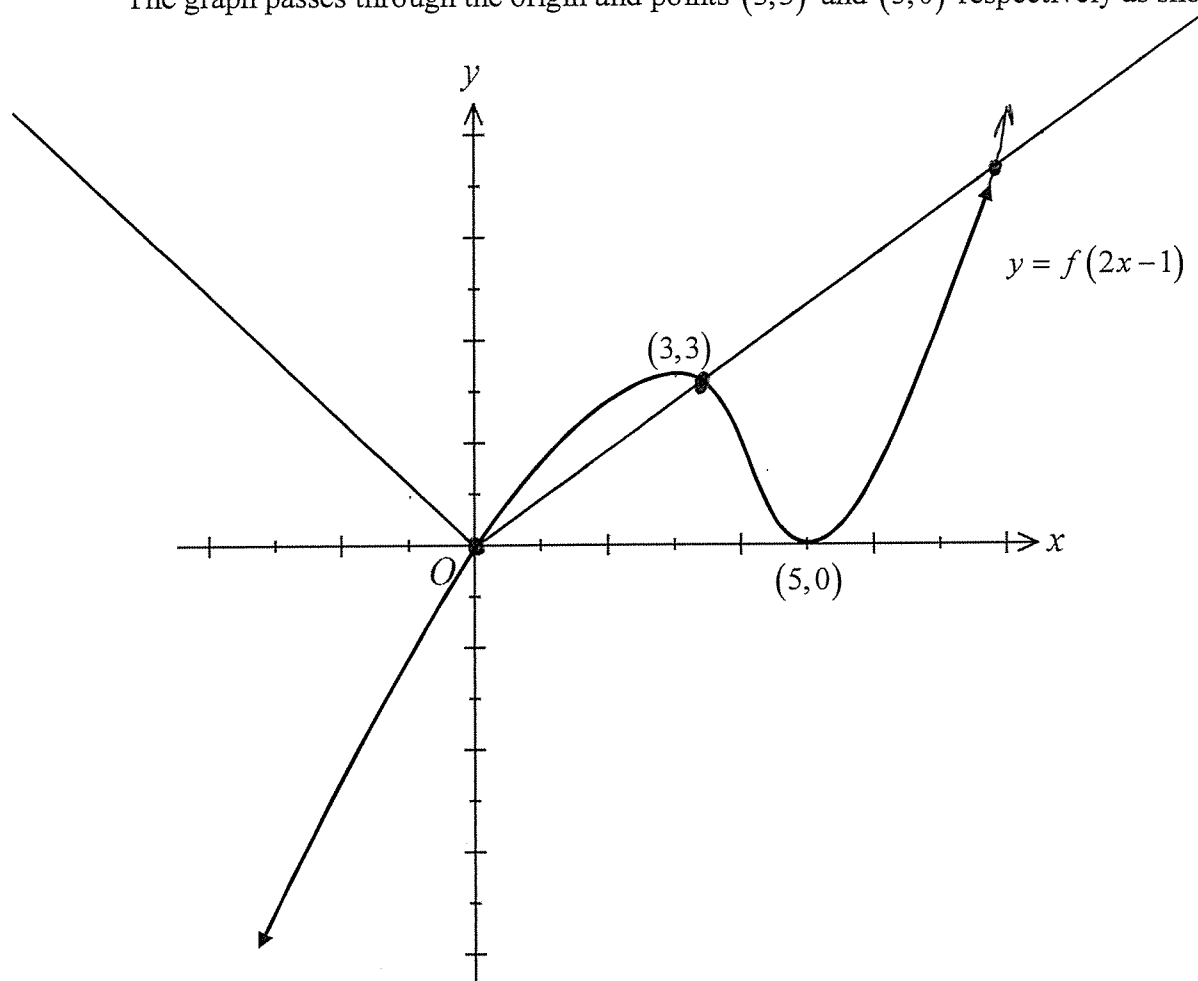
Question 9 (10 marks)

- (a) The graph of the curve $y = f(2x-1)$ is shown below.

$$= f\left(2\left(x-\frac{1}{2}\right)\right)$$

The graph passes through the origin and points $(3,3)$ and $(5,0)$ respectively as shown.

2



By using the graph, or otherwise:

- (a) Find the number of solutions to the equation $f(2x-1) - |x| = 0$.

1

$f(2x-1) = |x|$
three solutions

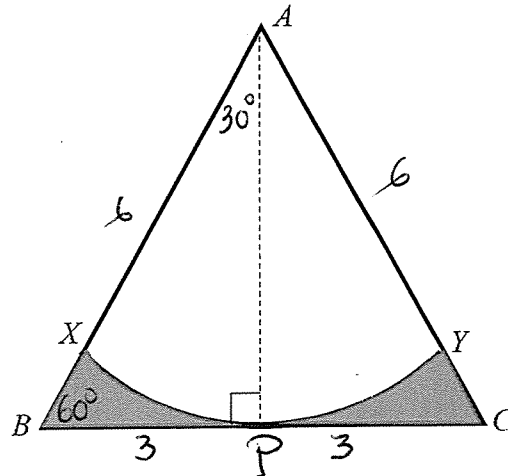
- (b) Solve $f(x) = 0$ two solutions $f(2x-1) = 0$ at $x = 1, 5$
 $f(2 \times 5 - 1) = f(9) = 0$
 $f(2 \times 0 - 1) = f(-1) = 0$
 $x = -1, 9$

Question 9 continues on Page 18

Question 9 (continued)

- (b) $\triangle ABC$ is an equilateral triangle with sides of length 6 m.

An arc with centre A and BC as tangent cuts AB and AC at X and Y respectively.



- (i) Show that the radius of the arc is $3\sqrt{3}$ m.

1

- (ii) Find the exact area of the shaded region.

3

$$\sin 60^\circ = \frac{AP}{6}$$

$$\frac{\sqrt{3}}{2} = \frac{AP}{6} \Rightarrow AP = \frac{6\sqrt{3}}{2} = 3\sqrt{3} \text{ m}$$

$$\text{ii) shaded A} = \text{Area } \triangle - \text{Area sector} \\ = \frac{6 \times 3\sqrt{3}}{2} - \frac{1}{2} \times \frac{\pi}{3} \times (3\sqrt{3})^2$$

$$= 9\sqrt{3} - \frac{1}{2} \times \frac{\pi}{3} \times 9 \times 3$$

$$= 9\sqrt{3} - \frac{9\pi}{2} \text{ or equivalent}$$

$$\text{OR shaded A} = \frac{6 \times 6 \times \sin \frac{\pi}{3}}{2} - \frac{1}{2} \times (3\sqrt{3})^2 \times \frac{\pi}{3}$$

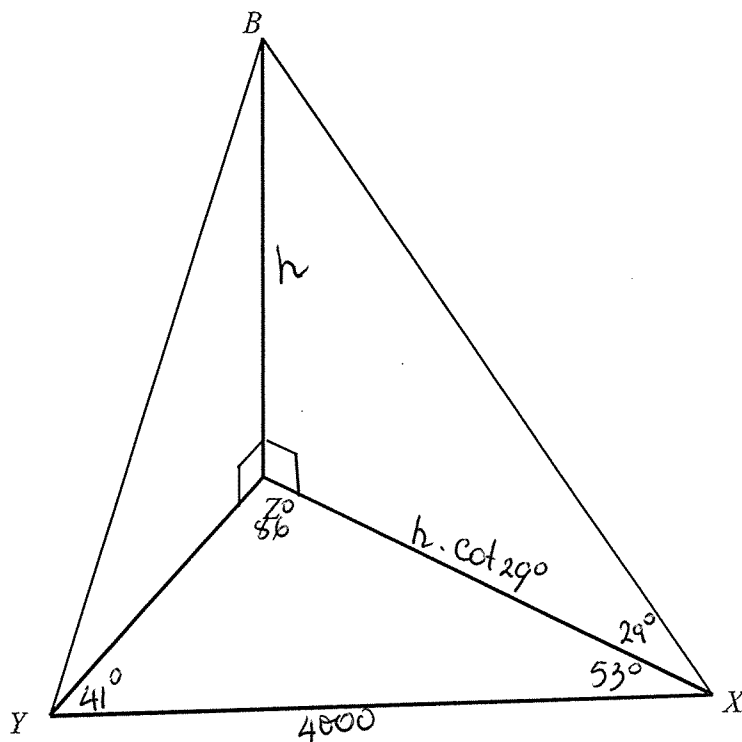
$$= 18 \times \frac{\sqrt{3}}{2} - \frac{9\pi}{2}$$

$$= 9\sqrt{3} - \frac{9\pi}{2}$$

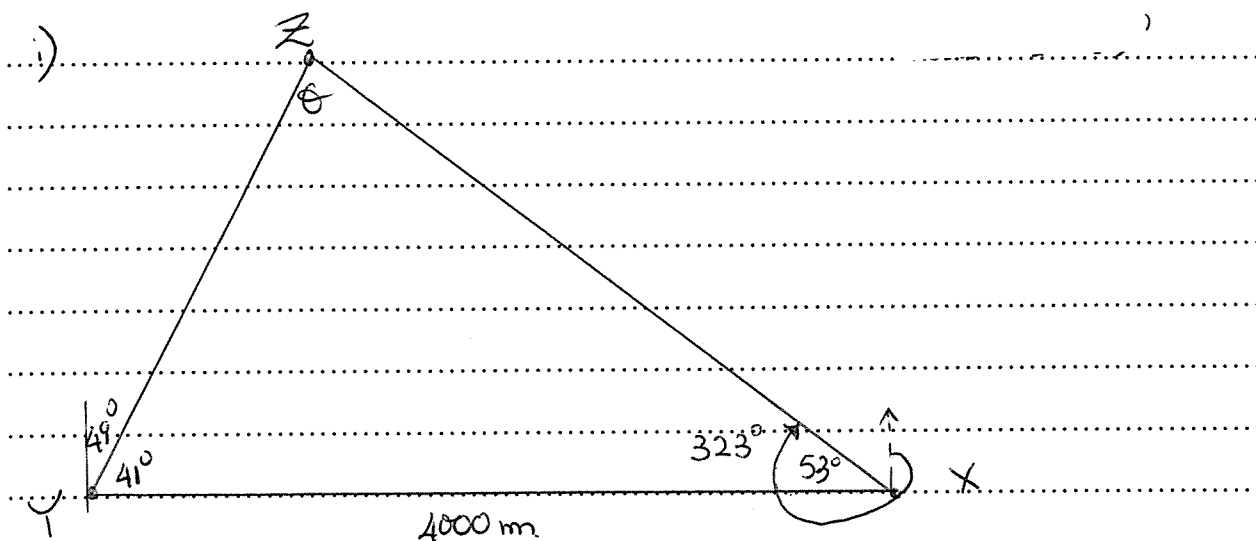
Question 9 continues on Page 19

Question 9 (continued)

- (c) The angle of elevation of a hot air balloon B from a point on the ground X is 29° and the bearing is 323° .
From point Y 4000 m due west of X , the bearing of the balloon is 049° .



- (i) Draw $\triangle XYZ$ showing the bearings. 1
(ii) Explain why $\angle YZX = 86^\circ$. 1
(iii) Find the altitude of the balloon. 2



- ii) let $\angle YZX = \theta$
 $\therefore \theta = 180^\circ - 53^\circ - 41^\circ = 86^\circ$

iii) let BZ be h

In $\Delta B X Z$

$$\tan 29^\circ = \frac{h}{XZ}$$

$$XZ = \frac{h}{\tan 29^\circ} = h \cot 29^\circ$$

In $\Delta X Y Z$:

Using Sine Rule

$$\frac{h \cot 29^\circ}{\sin 41^\circ} = \frac{4000}{\sin 86^\circ}$$

$$h = \frac{4000 \times \sin 41^\circ}{\sin 86^\circ \times \cot 29^\circ}$$

$$h = \frac{4000 \times \sin 41^\circ \times \tan 29^\circ}{\sin 86^\circ}$$

$$= 1458.2 \text{ m}$$

$$\approx 1459 \text{ m}$$

OR

$$\frac{h / \tan 29^\circ}{\sin 41^\circ} = \frac{4000}{\sin 86^\circ}$$

$$\frac{h}{\tan 29^\circ \sin 41^\circ} = \frac{4000}{\sin 86^\circ}$$

$$h = \frac{4000 \cdot \sin 41^\circ \cdot \tan 29^\circ}{\sin 86^\circ}$$

$$= 1459 \text{ m}$$

End of Paper